# Monthly Intelligence Report

## Edition 003 — The California Seismic Corridor

Where the energy transition is outrunning the grid. edition · June 2026 · SSI v4.0.2

CONTENTS



SECTION A

## Executive Summary

SUBSTATIONS SCORED

**45,003**

342 HV · 2,808 MV

GRID LINES MAPPED

**22,765**

87,000 km coverage

MC SIMULATIONS

**35.3 M**

10,000 per substation

PUBLIC DATA SOURCES

**30+**

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 65,300+ source data points per run, executes 35.3 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 794 million arithmetic operations — producing 209,722 output data points with full uncertainty quantification across 45,003 substations and 22,765 grid lines spanning 105,000 km.

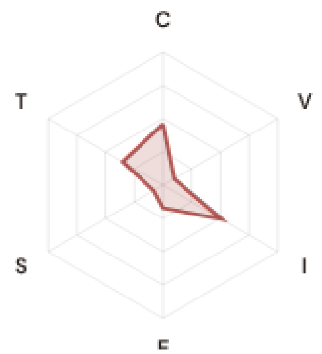
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering the US's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

## Substation in Focus

### TAP138880

Kansas, Kansas · 69 kV

| 0.316 | Medium | 0.128 | 29th |
| --- | --- | --- | --- |
| R_FINAL | CLASSIFICATION | CI WIDTH | FLEET PERCENTILE |




COMPONENTS

**C** Continuity: 0.461 / 0.30 (154%)
**I** Infrastructure: 0.512 / 0.25 (205%)
**S** Saturation: 0.081 / 0.20 (41%)
**V** Voltage: 0.097 / 0.10 (97%)
**E** Economic: 0.172 / 0.10 (172%)
**T** Transition: 0.351 / 0.05 (702%)

MODIFIERS

**R3 Consequence:** 0.955 ↓ dampens
**R4 Graph Criticality:** 0.986 ↓ dampens
**R6a Restoration:** 1.026 ↑ amplifies
**R6b Seismic:** 1.006 → neutral
**R7 Digital Readiness:** 0.999 → neutral

This month's spotlight — automatically selected for June 2026 — is **TAP138880** in Kansas, Kansas. With an R_final of 0.316 (Medium band, 29th fleet percentile), its risk profile is dominated by the T (Transition) component at 702% of its weight ceiling, followed by I (Infrastructure) at 205%. Modifiers collectively shift R_base by 3.5%%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

# Cyber & Economic Exposure Monitor

**Why this section exists:** Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that the US's Digital Strategy / Net Zero Strategy planning requires.

CYBER HIGH EXPOSURE
**0**
0.0% of fleet

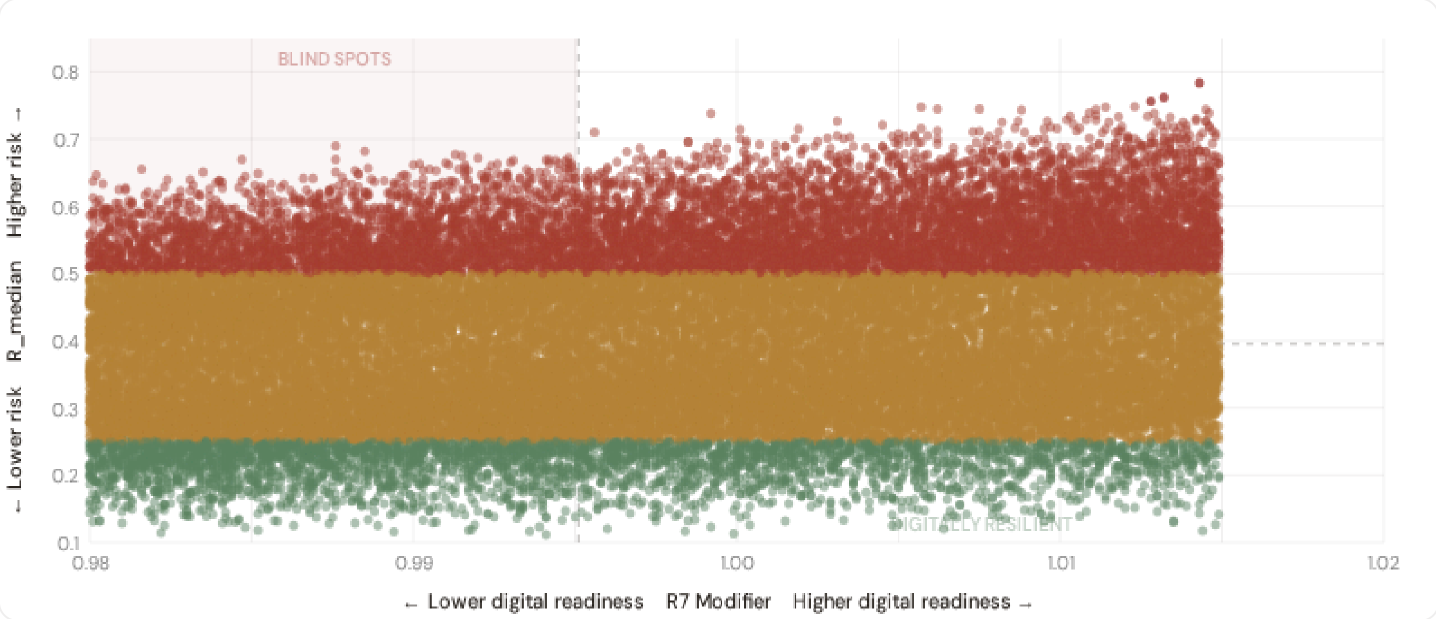
BLIND SPOTS
**3792**
High/Critical + R7 < median

AVG R7 MODIFIER
**0.9950**
Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS
**11,280**
25.1% · R3 ≥ 0.97

## B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.




● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **3792 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 0.9951). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Florida (307), Georgia (284), Indiana (264)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

## B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.

### 🏭 Capital-Intensive / High-Consequence

Est. VoLL: $15–30/kWh (USD)

Capital nodes — highest economic loss per interruption (semiconductor fabs, data centers, refineries, hospitals)

**11,280** substations (25.1%) High/Critical: **2280** (20.2%) Avg R: **0.391** Avg E: **0.128** / 0.10

Low 1339 Med 7661 High 2279 Crit 1

### 🔧 Industrial / Medium-Large Enterprise

Est. VoLL: $8–15/kWh (USD)

Industrial belt — significant continuity requirements (manufacturing, logistics hubs)

**11,319** substations (25.2%) High/Critical: **2591** (22.9%) Avg R: **0.400** Avg E: **0.130** / 0.10

Low 1169 Med 7559 High 2590 Crit 1

### 🏪 Commercial / SME-Dense

Est. VoLL: $3–8/kWh (USD)

Service economy + retail — moderate individual impact (commercial districts, mixed-use)

**11,129** substations (24.7%) High/Critical: **2598** (23.3%) Avg R: **0.400** Avg E: **0.128** / 0.10

Low 1199 Med 7332 High 2598 Crit 0

### 🌾 Light / Agricultural / Rural

Est. VoLL: $1–3/kWh (USD)

Sparse activity, agricultural land — lower VoLL (residential, agricultural)

**11,275** substations (25.1%) High/Critical: **3041** (27.0%) Avg R: **0.410** Avg E: **0.129** / 0.10

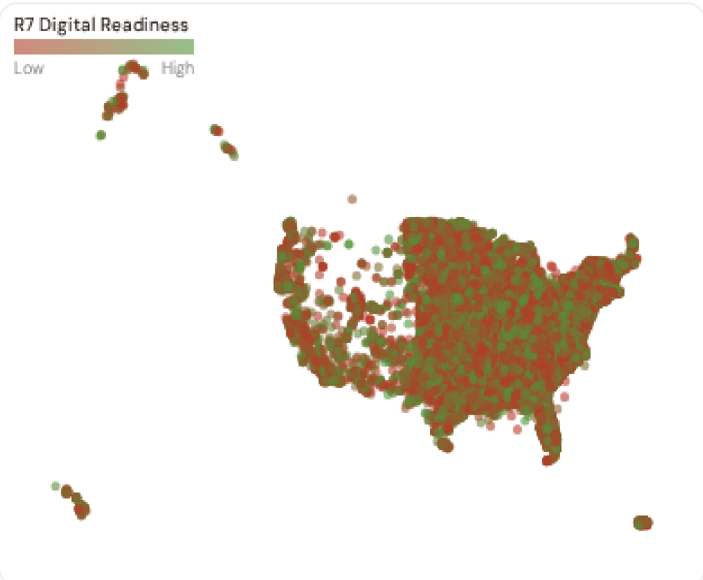
Low 1013 Med 7221 High 3040 Crit 1

**Value of Lost Load (VoLL) context:** ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **2280 substations** combine capital-intensive economic fabric (R3 ≥ 0.97) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **North Carolina (686), Alabama (676), Kentucky (537)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 11.7%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

## B.3 County Digital Readiness Ranking

Countys ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify counties combining digital exposure with high grid stress.




WORST DIGITAL READINESS — TOP 15 REGION REGIONS

Longer bar = higher R7 = better digital infrastructure.

| | | |
|---|---|---|
| 1 | New Hampshire | 0.9922 |
| 2 | Connecticut ⚠ high R | 0.9935 |
| 3 | Arizona | 0.9937 |
| 4 | Alaska | 0.9939 |
| 5 | Washington ⚠ high R | 0.9940 |
| 6 | Utah ⚠ high R | 0.9944 |
| 7 | Illinois ⚠ high R | 0.9945 |
| 8 | Hawaii ⚠ high R | 0.9946 |
| 9 | Mississippi | 0.9946 |
| 10 | Arkansas | 0.9946 |
| 11 | Pennsylvania ⚠ high R | 0.9947 |
| 12 | Indiana ⚠ high R | 0.9947 |

| | | |
|---|---|---|
| 13 | Delaware ⚠ high R | 0.9948 |
| 14 | Florida ⚠ high R | 0.9948 |
| 15 | South Dakota | 0.9948 |

region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **New Hampshire** has the lowest average R7 (0.9922), while **District of Columbia** leads at 1.0008 — a spread of 0.0086 across the R7 range. **14 region regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

### B.4 Monthly Pulse

**Inaugural edition — Baseline Snapshot (June 2026).** This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9950, median = 0.9951; 0 substations classified HIGH cyber-exposure; 3792 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

## Data Refresh & Changelog

The SSI v4.0.2 draws on **40 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

40

95 variables

WEEKLY / MONTHLY

4

High-frequency feeds

QUARTERLY / ANNUAL

21

Regular refresh cycle

STATIC / DERIVED

8

Standards + scenarios

## Data Source Registry — June 2026

| | | | | |
|---|---|---|---|---|
| OSM | OSM Power Infrastructure | WEEKLY | Node/edge | 3 vars |
| BLS-LM | BLS Labour Market Statistics | MONTHLY | County | 1 var |
| DOE | DOE/EIA (Department of Energy) | QUARTERLY | County | 5 vars |
| EIA | EIA (Energy Information Administration) | ANNUAL | County | 8 vars |
| Census | Census Bureau (US Census Bureau) | ANNUAL | County | 8 vars |
| NERC | NERC (North American Electric Reliability Corp) | ANNUAL | Regional | 2 vars |
| FEMA | FEMA / EPA (Federal Emergency Management Agency) | ANNUAL | County | 4 vars |
| EEA | EEA Air Quality e-Reporting | ANNUAL | ~1 km + station | 3 vars |
| BLS | BLS (Bureau of Labor Statistics) | ANNUAL | County | 3 vars |
| NTIA | NTIA Broadband Map | ANNUAL | County | 2 vars |
| CISA | CISA (Cybersecurity and Infrastructure Security Agency) | ANNUAL | National | 2 vars |
| FEMA-FR | FEMA Flood Risk / Hazard Mitigation | ANNUAL | County | 2 vars |
| NRC | NRC (Nuclear Regulatory Commission) | ANNUAL | Site-level | 1 var |
| BEA | BEA (Bureau of Economic Analysis) | ANNUAL | County | 3 vars |
| EIA-SS | EIA Statistical Series | ANNUAL | State/County | 2 vars |
| FRA | FRA (Federal Railroad Administration) | ANNUAL | County | 1 var |
| DOT | DOT/FHWA (Dept of Transportation) | ANNUAL | County | 1 var |
| FERC | FERC Monitoring Report | ANNUAL | Regional | 2 vars |
| PUC | State PUCs (Public Utility Commissions) | ANNUAL | Service Territory | 2 vars |
| CESER | DOE CESER (Cybersecurity, Energy Security, Emergency Response) | ANNUAL | Regional | 2 vars |
| Eurostat | Eurostat Energy Statistics | ANNUAL | EU Regional | 3 vars |
| JRC | JRC European Commission | ANNUAL | European | 2 vars |

| | | | | |
|---|---|---|---|---|
| USGS | USGS (US Geological Survey) | STATIC | County | 3 vars |
| CDS | Copernicus CDS / ERA5 | STATIC | 0.25° (~25 km) | 4 vars |
| DIM | Dimovski et al. (2025) | STATIC | County | 3 vars |
| IEEE | IEEE C57.91 / IEC 60076 | STATIC | Asset-level | 16 vars |
| GFZ | GFZ German Research Centre | STATIC | ~5 km | 2 vars |
| USGS-SH | USGS Seismic Hazard Assessment | STATIC | County | 1 var |
| Census-GEO | Census Bureau Geography Registry | STATIC | County | 1 var |
| USGS-HZ | USGS Hazard Assessments | STATIC | ~5 km | 2 vars |
| ISO9223 | ISO 9223 Corrosion | DERIVED | County | 1 var |
| EIA-ISO | EIA/ISO-RTE (PJM/CAISO/MISO) — Data Portal | HOURLY | Balancing Authority | 3 vars |
| OM | Open-Meteo Historical Weather | HOURLY | ~1 km | 3 vars |
| NOAA | NOAA/NWS (National Oceanic and Atmospheric Admin) | HOURLY | ~1 km | 2 vars |
| NOAA-NWS | NOAA/NWS (National Weather Service) | HOURLY | Station | 3 vars |
| ENTSO-E | ENTSO-E Transparency | HOURLY | Cross-border | 2 vars |

## C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

**Weekly** (2 sources) — OSM diffs + weather reanalysis

OSM Overpass · Open-Meteo/ERA5

**Monthly** (2 sources) — Grid load + procurement data

DOE Registry · EIA-ISO Transparency

**Quarterly** (1 sources) — DER registry + business + macro

DOE

**Annual** (19 sources) — Core grid quality + demographics

EIA · Census · FEMA · BLS-LM · DOE · EIA-SS · FERC · EPA · Eurostat · BLS · NTIA · CISA · NOAA-NWS · USGS · NRC · FRA · BEA · PUC · CESER

**Static** (8 sources) — Standards + scenarios + seismic

Dimovski · IEEE/IEC/CIGRÉ · Copernicus CMIP6 · GFZ · USGS · ISO 9223 · Census/OSM · USGS Seismic Hazard

## C.2 Fleet Score Snapshot

FLEET SIZE

**45,003**

substations scored

MEDIAN R

**0.395**

P5=0.220 · P95=0.598

HIGH + CRITICAL

**10510**

23.4% of fleet

HIGH CONFIDENCE

**100%**

44981 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 4720 (10.5%) Medium 29773 (66.2%) High 10507 (23.3%) Critical 3 (0.0%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **45,003 substation scores remain unchanged** this edition. The fleet median R of 0.395 (mean 0.400) reflects a distribution skewed toward the lower bands, with 10.5% in Low and 66.2% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

## C.3 Changelog — v3.4 → v4.0.2

18 changes tracked across P0, P1, and P2 releases

| | | | |
|---|---|---|---|
| F1 | NEW | New T component — Energy Transition Exposure (T1) | §2, §4 |
| F2 | NEW | New R6a — Restoration Speed Modifier (CAIDI-based) | §5 |
| F3 | ENHANCED | R3 enhanced — Energy Poverty Vulnerability (V_socio) | §5 |
| F4 | ENHANCED | R4 enhanced — Graph-theoretic betweenness + bridge detection | §5 |
| F5 | ENHANCED | R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5) | §5 |
| F6 | ENHANCED | R7 Digital Readiness — County-level NTIA Broadband model | §5 |
| L1 | NEW | New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeological | §2, §8 |
| L2 | ENHANCED | E2 Innovation Enrichment — BEA + economic structure | §6 |
| L3 | ENHANCED | V_socio Fiscal Enrichment — Census + BLS + energy price | §5 |
| L4 | ENHANCED | R3 Demographic Shift Amplifier — Census net migration | §5 |
| L5 | DATA | 95/95 variables operational (100%). 40 data sources total. | §8, §12 |
| G1 | DATA | DOE upgraded to quarterly registry — County-level DER registry | §12 |
| G2 | DATA | USGS upgraded to live API — County-level seismic data | §12 |

| | | | |
|---|---|---|---|
| G3 | DATA | OSM upgraded to Overpass API — 45,003 real substations (US) | §12 |
| G4 | NEW | R6b Seismic Hazard modifier — USGS Seismic Hazard classification | §5 |
| G5 | NEW | Network & Topology layer (H): 7 variables — EIA Grid Plan, seismic analysis | §12 |
| G6 | NEW | Output Scores layer (I): 7 variables — risk_score, ETTC, stationary probs | §12 |
| G7 | REGIONAL | US Data Sources — EIA, Census, USGS, FEMA, NRC, DOT, NERC, state PUCs | §12 |

SECTION D — MONTHLY DEEP-DIVE

# The California Seismic Corridor: Where Seismic Risk Meets Energy Transition Pressure

**Deep-dive rotation:** Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** California seismic corridor · **002** DER saturation hotspots (East Anglia/West Midlands) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan-Rural economic impact disparity · **005** Nuclear fleet dependency and phase-out risk · **006** Infrastructure age vs smart meter rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** County Grid Health Cards · **010** Academic validation and peer feedback · **011** European replication feasibility · **012** Year in review.

## D.1 Why the California, Why Now

CALIFORNIA SUBSTATIONS

**372**

47 EHV · 325 HV

HIGH BAND

**252**

67.7% of corridor

CORRIDOR MEDIAN R

**0.553**

Fleet median: 0.395

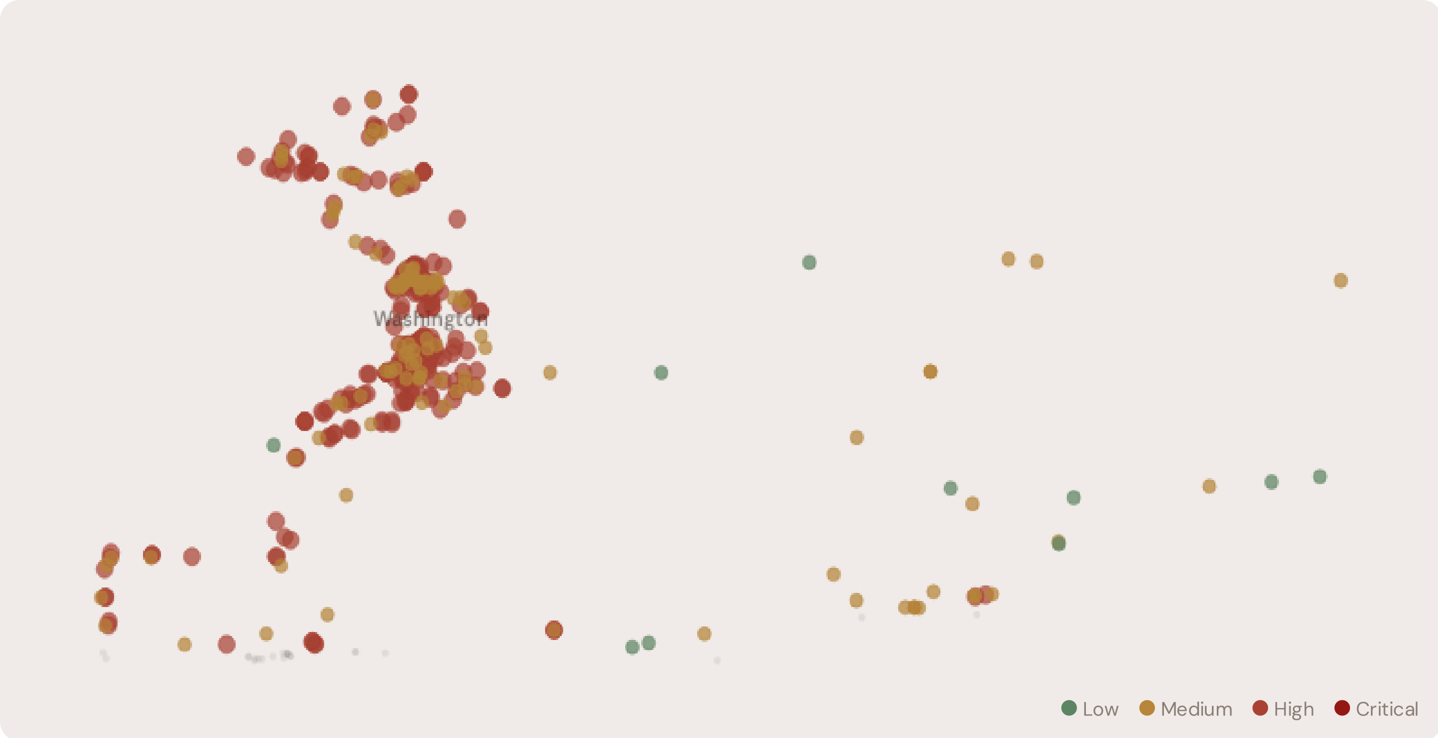
WORST COUNTY

**Washington**

Avg R: 0.522 · 372 substations

The Washington hosts **372 substations** across 1 region regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **252 substations (67.7%)** are classified High, with only **10** in the Low band. 84% of corridor substations score above the national fleet median of 0.395. The corridor median R of 0.553 reflects the local risk profile. The corridor concentrates the country's industrial heartland. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under national digitalisation timelines.

## D.2 Corridor Map and Scoring




| SUBSTATION | COUNTY | R_FINAL | BAND | C | V | I | E | S | T | ALERT |
|---|---|---|---|---|---|---|---|---|---|---|
| TAP209806 | Washington | 0.733 | High | 0.898 | 0.280 | 0.643 | 0.403 | 0.928 | 0.700 | C, S, T |
| PICNIC POINT | Washington | 0.729 | High | 0.882 | 0.415 | 0.626 | 0.454 | 0.896 | 0.613 | C, S |
| KENMORE | Washington | 0.727 | High | 0.873 | 0.462 | 0.665 | 0.483 | 0.920 | 0.664 | C, I, S, T |
| UNKNOWN202934 | Washington | 0.701 | High | 0.813 | 0.297 | 0.584 | 0.379 | 0.916 | 0.727 | C, S, T |
| PRESIDENT PARK | Washington | 0.697 | High | 0.908 | 0.359 | 0.609 | 0.485 | 0.908 | 0.718 | C, S, T |
| SNOHOMISH PUD | Washington | 0.696 | High | 0.896 | 0.274 | 0.626 | 0.499 | 0.908 | 0.573 | C, S |
| NORLUM | Washington | 0.695 | High | 0.750 | 0.364 | 0.630 | 0.506 | 0.811 | 0.731 | C, S, T |
| TUKWILA STATION | Washington | 0.694 | High | 0.816 | 0.297 | 0.636 | 0.308 | 0.892 | 0.510 | C, S |
| TAP206097 | Washington | 0.691 | High | 0.827 | 0.374 | 0.663 | 0.515 | 0.857 | 0.624 | C, I, S |
| OLYMPIC VAIL PIPELINE | Washington | 0.690 | High | 0.764 | 0.449 | 0.571 | 0.524 | 0.814 | 0.675 | C, S, T |

*… 362 additional substations — explore full corridor in Digital Twin →*

## D.3 Component Analysis

COMPONENT AVERAGES — WASHINGTON VS FLEET

MODIFIER AVERAGES — WASHINGTON VS FLEET

| | |
|---|---|
| C — Continuity | 0.626 vs 0.505 (+24%) |
| I — Infrastructure | 0.482 vs 0.552 (-13%) |
| S — Saturation | 0.725 vs 0.354 (+105%) |
| V — Voltage | 0.311 vs 0.261 (+19%) |
| E — Economic | 0.428 vs 0.129 (+232%) |
| T — Transition | 0.548 vs 0.447 (+23%) |

| | | |
|---|---|---|
| R3 Consequence | 0.955 | fleet 0.955 ↓ |
| R4 Graph Crit. | 0.979 | fleet 0.980 ↓ |
| R6a Restoration | 1.019 | fleet 1.020 ↑ |
| R6b Seismic | 0.999 | fleet 1.000 → |
| R7 Digital Read. | 0.994 | fleet 0.995 → |

The dominant risk driver across the Washington corridor is **T (Transition)**, averaging 0.548 against a fleet average of 0.447 — occupying 1096% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.428 vs fleet 0.129 (428% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.955 (fleet: 0.955), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 0.999, capturing the local seismic exposure — a dimension invisible to every competing framework.

## D.4 County Breakdown

Loading county breakdown...

## D.5 Implications

**29 Washington substations** score R ≥ 0.650, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the Washington sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

# American Context Card

> **How this works:** Each edition includes one American Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the California corridor analysis hinges on the US's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

### Unplanned SAIDI (min/customer/year) — US vs International Peers

Longer bar = lower SAIDI = better reliability. Values in min/customer/year (excl. major events).

| | |
|---|---|
| 🇩🇪 Germany | 11.7 |
| 🇨🇭 Switzerland | 15 |
| 🇳🇱 Netherlands | 23.9 |
| 🇬🇧 United Kingdom | 36 |
| 🇮🇹 Italy | 43 |
| **🇺🇸 US (national)** | **120** |
| ↳ Urban | 85 |
| ↳ Rural | 240 |
| ↳ Best (DC/NE) | 37 |
| 🇪🇸 Spain | 51 |
| 🇵🇱 Poland | 95 |
| 🇷🇴 Romania | 290 |

Source: EIA Form 861 (2023) for US national/urban/rural/state; CEER 7th Benchmarking Report (2022) for EU peers; Bundesnetzagentur (2024) for Germany. US values exclude major event days (IEEE 1366 standard). · See §16 Peer Benchmarking for full methodology. · Updated annually.

**This month's key insight:** The Washington corridor deep-dive shows **346 substations** (of 372) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.626 is **1.2× the fleet average** of 0.505. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

## Regional Risk Ranking — All 52 States

Where does the California sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.

| | | |
|---|---|---|
| #1 | **Washington** | 372 subs · avg R = 0.522 · H:252 |
| #2 | New Jersey | 561 subs · avg R = 0.489 · H:239 |
| #3 | Maryland | 664 subs · avg R = 0.486 · H:296 |
| #4 | Massachusetts | 483 subs · avg R = 0.474 · H:184 |
| #5 | Connecticut | 255 subs · avg R = 0.467 · H:108 |

#6 Indiana — 1340 subs · avg R = 0.465 · H:555

#7 Florida — 1719 subs · avg R = 0.457 · H:732

#8 Oregon — 226 subs · avg R = 0.451 · H:84

#9 Pennsylvania — 790 subs · avg R = 0.448 · H:292

#10 Texas — 1652 subs · avg R = 0.448 · H:658

#11 West Virginia — 594 subs · avg R = 0.448 · H:216

#12 Illinois — 1472 subs · avg R = 0.446 · H:518

#13 Georgia — 1952 subs · avg R = 0.432 · H:684

#14 Delaware — 154 subs · avg R = 0.427 · H:53

#15 District of Columbia — 23 subs · avg R = 0.422 · H:6

#16 Hawaii — 79 subs · avg R = 0.421 · H:18

#17 California — 774 subs · avg R = 0.421 · H:244

#18 Rhode Island — 99 subs · avg R = 0.416 · H:31

#19 New York — 1594 subs · avg R = 0.413 · H:420 C:3

#20 Michigan — 1243 subs · avg R = 0.413 · H:343

#21 Minnesota — 2127 subs · avg R = 0.407 · H:554

#22 Utah — 194 subs · avg R = 0.402 · H:24

#23 Ohio — 1516 subs · avg R = 0.401 · H:379

#24 Idaho — 13 subs · avg R = 0.400 · H:0

#25 Missouri — 1541 subs · avg R = 0.395 · H:366

#26 Arizona — 197 subs · avg R = 0.384 · H:43

#27 South Carolina — 1597 subs · avg R = 0.382 · H:325

#28 Wisconsin — 1408 subs · avg R = 0.381 · H:238

#29 Alaska — 129 subs · avg R = 0.381 · H:15

#30 North Carolina — 2720 subs · avg R = 0.380 · H:467

#31 Colorado — 36 subs · avg R = 0.378 · H:6

#32 New Hampshire — 80 subs · avg R = 0.377 · H:15

#33 Kentucky — 2192 subs · avg R = 0.377 · H:376

#34 Arkansas — 1028 subs · avg R = 0.377 · H:162

#35 Nevada — 101 subs · avg R = 0.377 · H:20

#36 Montana — 21 subs · avg R = 0.373 · H:1

#37 Alabama — 2549 subs · avg R = 0.371 · H:386

#38 Iowa — 1197 subs · avg R = 0.369 · H:149

#39 Louisiana — 1126 subs · avg R = 0.369 · H:141

#40 Virginia — 909 subs · avg R = 0.365 · H:123

#41 New Mexico — 273 subs · avg R = 0.365 · H:20

#42 Tennessee — 1522 subs · avg R = 0.360 · H:204

#43 Oklahoma — 2031 subs · avg R = 0.359 · H:171

#44 Kansas — 746 subs · avg R = 0.359 · H:82

#45 Wyoming — 16 subs · avg R = 0.359 · H:1

#46 Mississippi — 866 subs · avg R = 0.355 · H:99

#47 Maine — 172 subs · avg R = 0.353 · H:16

#48 Vermont — 82 subs · avg R = 0.352 · H:10

#49 North Dakota — 798 subs · avg R = 0.352 · H:54

#50 Nebraska — 895 subs · avg R = 0.349 · H:77

#51 South Dakota — 751 subs · avg R = 0.344 · H:43

#52 Unassigned — 124 subs · avg R = 0.321 · H:7

The Washington ranks **#1 of 52 region regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are Washington, New Jersey, Maryland, while the three most resilient are Unassigned, South Dakota, Nebraska. 24 of 52 score above the fleet average of 0.400. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

## SSI Enhanced Neural Network — Monthly Topic

**From grid intelligence to BESS valuation platform.** The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the ISO/RTO and utility, and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.

Layer 1 — Data | Layer 2 — Valuation | **Layer 3 — Neural Net** | Layer 4 — Coverage | Layer 5 — Platform

| | | | | |
|---|---|---|---|---|
| SSI v4.0 component taxonomy, normalisation, MC engine | DSO/TSO + Community value stacks, TCO, real options | GBT+DNN ensemble, transfer learning, SHAP | Country prioritisation, DCI, registry, calibration | Products, API, revenue model, unit economics |

LAYER 3 §0.5.6 · June 2026

### Explainability & SHAP Values

*Clients need to understand not just the prediction but WHY a substation ranks highly*

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

**KEY CONCEPTS**

- → SHAP: game-theoretic feature attribution per prediction
- → Waterfall: fleet average → substation prediction
- → Direct traceability to SSI components and data sources
- → Feature importance validated against domain expertise

**LIVE SSI DATA CONNECTION**

The training set comprises 45,003 substations with complete feature vectors. Average CI_width of 0.147 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 4720 Low, 29773 Medium, 10507 High, 3 Critical — ensures balanced training across all risk bands.

**Coming next month:** LAYER 3 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

## Looking Ahead

NEXT MONTH — EDITION 02

### Mountain Region

Deep-dive into Region 10's grid risk profile — substation map, component analysis, region regions breakdown. European Context Card: Light-rural tier — limited substation density.

NEXT DATA REFRESH

### Q3 2026 — 1 Quarterly Sources

**DOE** are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **EIA (Energy Information Administration)** (8 variables → C1–C4 (SAIDI/SAIFI/CAIDI), quality regulation).

FLEET SPOTLIGHT

### Washington: Highest Risk Concentration

Washington has the highest concentration of High/Critical substations: **252 of 372 (68%)** are in the upper risk bands — avg R = 0.522. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Region 10**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

**Feedback welcome.** The SSI Index is designed to evolve through stakeholder dialogue. If you represent a US utility, FERC, NERC, an RTO/ISO (CAISO, ERCOT, MISO, PJM, ISO-NE, NYISO, SPP), a DOE national laboratory, a state public utility commission, an investor-owned or municipal utility, or an academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 003

Published 11 June 2026 · SSI Index v4.0.2 · 45,003 substations · 22,765 grid lines · 6 components · 20 metrics · 6 modifiers

10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5